INTERNATIONAL UNIVERSITY

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Educational Program: 6B06112 Data Science

AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

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Motivation and Aim

Motivation

Temperature changes in rocks affect stress, deformation, and wave propagation. These processes are relevant to geothermal systems, underground engineering, and geophysical interpretation.

Aim

Develop and justify an AI-guided framework for directional prediction of thermoelastic wave propagation in geological media under simplified effective-medium assumptions.

Scope

Research prototype for source–probe directional prediction, not a field-validated simulator.

Basalt – example of a target geological medium.

Temperature change
↓
Stress and deformation
↓
Thermoelastic wave
↓
Direction prediction

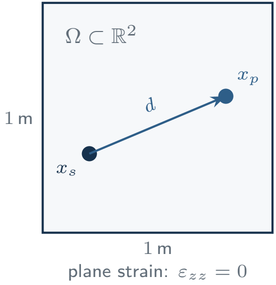
Threats to Validity

1. **Limited training data.** The COMSOL-derived pilot configuration is not a full calibrated geological database.
2. **Simplified 2D assumptions.** The study uses effective material parameters in a plane-strain setup.
3. **FNO normalization issue.** The current 2D adaptation is not yet physically reliable.
4. **Analytical reference is isotropic.** r/V_p is a useful sanity check, not a complete thermoelastic validation.

Geological Media We Model

Rock	ρ kg/m^3	V_P m/s	k W/(m K)	C_p J/(kg K)	α 10^{-6}/K
Granite	2650	5850	2.50	850	7.9
Basalt	3000	6550	1.95	1300	–
Sandstone	2725	5000	2.25	1000	11.6
Limestone	2675	5400	2.55	1050	8.0

Effective scalar parameters under a locally homogeneous isotropic approximation; field calibration is out of scope. Full 10-rock tables with E, ν, μ, K, porosity are in the backup slides.



2D plane-strain domain with source–probe geometry.

Coupled Thermoelastic System

Constitutive law (Duhamel–Neumann)

$$\sigma_{ij} = \lambda\,\delta_{ij}\,\varepsilon_{kk} + 2\mu\,\varepsilon_{ij} - \gamma\,\delta_{ij}\,\theta, \quad \gamma = 3K\alpha, \quad \theta = T - T_0$$

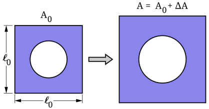
Equation of motion

$$\rho\,\frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i$$

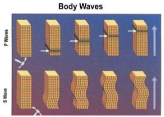
Coupled heat equation

$$\rho C_p\,\frac{\partial T}{\partial t} - k\,\nabla^2 T + \gamma T_0\,\frac{\partial}{\partial t}(\nabla\cdot u) = Q$$

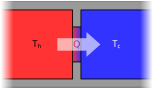
σ stress; ε strain; λ, μ, K moduli; $\gamma = 3K\alpha$ coupling; $\theta = T - T_0$; Q heat source.



Thermal expansion $\varepsilon_{\mathrm{th}} = \alpha\Delta T$



Elastic body waves (motion)



Heat flux $q = -k\nabla T$

2D Directional Prediction Task

Scope

Domain $\Omega \subset \mathbb{R}^2$, plane strain, linear isotropic thermoelasticity, small deformation. Primary fields are the temperature $T(x,t)$ and the in-plane displacement $u(x,t) = (u,v)$.

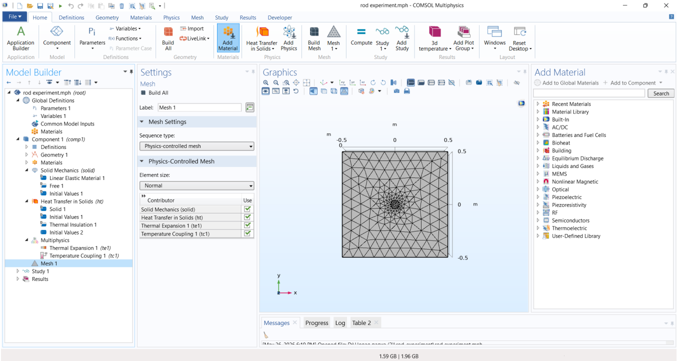
Directional query

$$d = x_p - x_s, \quad L = \|d\|, \quad \hat{d} = \frac{d}{L}$$

$$\phi = \operatorname{atan2}(d_y, d_x), \qquad \mathcal{F} : X \to \widehat{Y}$$

x_s, x_p source and probe positions [m]; L distance [m]; $\hat{d}$ unit direction; ϕ azimuth [rad]. $\mathcal{F}$ maps input features X (material, geometry, time) to the predicted response $\widehat{Y}$ (temperature, displacement, directional summary).

COMSOL-Based Synthetic Dataset



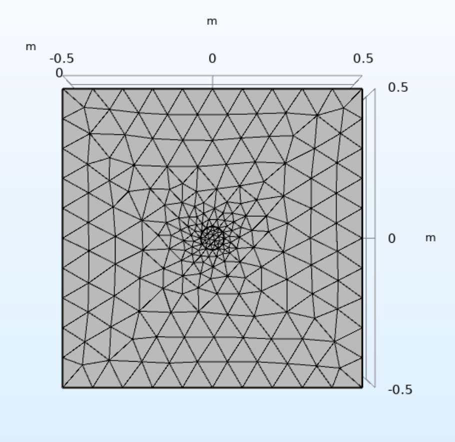
COMSOL simulation

The synthetic dataset was generated in COMSOL Multiphysics using a coupled thermoelastic model.

- 2D plane-strain geological domain
- Domain size: $1\,\mathrm{m} \times 1\,\mathrm{m}$
- Initial temperature: $293.15\,\mathrm{K}$
- Source temperature: $1500\,\mathrm{K}$
- Exported fields: T, (u, v), stress, strain, velocity

The exported COMSOL fields are used as synthetic reference data for comparing neural surrogate models.

COMSOL 2D Model and Mesh



Model domain

The numerical experiment represents a two-dimensional section of a geological medium with an internal heated source.

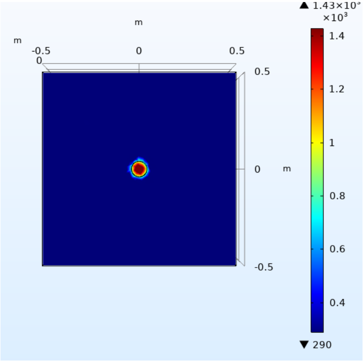
Mesh refinement

The mesh is refined near the source, where the strongest temperature, displacement, and stress gradients are expected.

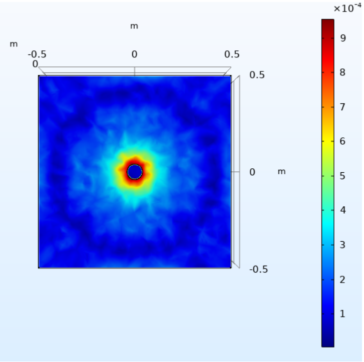
Role in the study

This mesh defines the spatial structure from which the COMSOL fields are exported for further preprocessing.

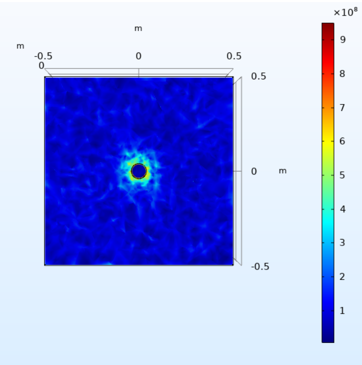
COMSOL Simulation Results



Local heating creates a compact temperature disturbance near the source.



Thermal expansion produces a displacement field around the heated region.



The stress field is concentrated near the localized thermal source.

Data Representation and Preprocessing

- COMSOL simulation results are exported as CSV files for each field and time step.
- The exported data are aligned into a common node-based representation.
- Dynamic fields are assembled into a spatio-temporal tensor:

$$\mathbf{S} \in \mathbb{R}^{N_t \times N_p \times F_d}$$

- Static material and geometric features are stored separately.
- All feature channels are normalized before model training:

$$\widetilde{x} = \frac{x - \mu}{\sigma + \varepsilon}$$

Here, N_t is the number of time steps, N_p is the number of spatial points, and F_d is the number of dynamic physical fields. The same processed dataset is then reused by coordinate-based, grid-based, and graph-based neural models.

Comparative Prototype Framework

One workflow for all model routes

The frontend, backend orchestrator, material catalog, and model services use one shared input–output contract for fair comparison.

- Same source–probe geometry for all routes
- Same effective geological material presets
- Same normalized response format: temperature, displacement, direction, travel time, and diagnostics

Shared Model Comparison Contract

Common interpretation

All routes are treated as surrogate modelling approaches for the same AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media task, not as complete numerical solvers.

Input

Material parameters, source–probe geometry, observation time, and processed field features.

Output

Estimated temperature, displacement components, travel time, and diagnostic response quantities.

Comparison basis

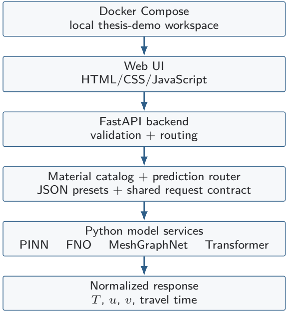
Same material presets, geometry contract, response format, and analytical travel-time sanity check.

Implementation Stack

Local thesis-demo stack

The prototype is packaged as a local Docker Compose workspace with a FastAPI orchestration backend, a native web frontend, and separate Python model services.

- **Frontend:** HTML, CSS, JavaScript interface for scenario setup.
- **Backend:** FastAPI validates requests and normalizes responses.
- **Services:** Python inference endpoints for PINN, FNO, MeshGraphNet, and Transformer routes.
- **Deployment:** Docker Compose starts the demo stack locally.



The backend does not replace the models; it orchestrates model-specific services and returns one comparable response format.

PINN: Physics-Informed Baseline

Role in the framework

PINN is the explicitly physics-informed route in the comparison.

- Uses supervised data and thermoelastic residuals.
- Predicts temperature and displacement fields.
- Serves as the physics-informed baseline for data-driven routes.

Interpretation

PINN is a neural surrogate whose loss encourages physical consistency. It is not an exact numerical solver.

Practical setup

Used in the simplified 2D source–probe prediction workflow.

PINN Input and Architecture

Public neural contract

$$[x, y, t, E, \nu, \rho, \alpha, k, C_p] \rightarrow [T, u, v]$$

Input layer
9 features

↓

Linear $9 \rightarrow 192$
Tanh

↓

$5\times$ hidden blocks
Linear $192 \rightarrow 192$ + Tanh

↓

Output layer
Linear $192 \rightarrow 3$

Implemented network. `MLP_PINN`, hidden dimension 192, depth 6, default activation `tanh`.

Input features

x, y, t	geometry and time
E, ν	elastic parameters
ρ	density
α, k, C_p	thermal parameters

Output features. $\hat{Y} = [\hat{T}, \hat{u}, \hat{v}]$. Temperature and two in-plane displacement components are predicted. Velocity, stress, and strain are derived through autograd during residual and velocity-consistency evaluation.

PINN Composite Loss and Residuals

Hybrid training objective

$$\mathcal{L} = \lambda_{\mathrm{sup}}\mathcal{L}_{\mathrm{sup}} + \lambda_{\mathrm{vel}}\mathcal{L}_{\mathrm{vel}} + \lambda_{\mathrm{wave}}\mathcal{L}_{\mathrm{wave}} + \lambda_{\mathrm{th}}\mathcal{L}_{\mathrm{th}}$$

- $\mathcal{L}_{\mathrm{sup}}$: supervised field loss for T, u, v.
- $\mathcal{L}_{\mathrm{vel}}$: velocity consistency from time derivatives.
- $\mathcal{L}_{\mathrm{wave}}$: elastic wave residual.
- $\mathcal{L}_{\mathrm{th}}$: coupled thermal residual.

Current limitation

Dedicated boundary-condition and initial-condition losses are not yet enforced.

In the implemented 2D mode, strain, stress, wave residual, and coupled thermal residual are computed from automatic derivatives.

Fourier Neural Operator

Operator view

$$\mathcal{G}_\theta : \mathcal{A} \to \mathcal{U}, \qquad \hat{s} = \mathcal{G}_\theta(a)$$

Input tensor

$$A \in \mathbb{R}^{B \times C_{in} \times N_x \times N_y}$$

Output tensor

$$\hat{S} \in \mathbb{R}^{B \times C_{out} \times N_x \times N_y}$$

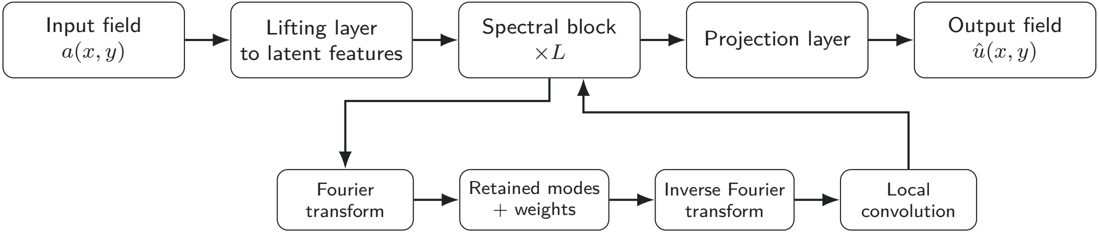
a – input field; $\hat{s}$ – predicted thermoelastic field; $\mathcal{A}$ and $\mathcal{U}$ – input and output field spaces; θ – trainable neural-operator parameters.

- B is batch size.
- N_x, N_y are grid sizes.
- C_{in} is the number of input channels.
- C_{out} is controlled by `out_channels`.

For the three-channel thermoelastic setup:

$$C_{out} = 3, \qquad \hat{s} = [\hat{T}, \hat{u}, \hat{v}].$$

FNO Pipeline



FNO maps an input field to an output field by combining global spectral interactions and local spatial corrections.

FNO Spectral Block

Layer update

$$v_{l+1} = \sigma\left(\mathcal{S}_l(v_l) + \mathcal{C}_l(v_l)\right)$$

where v_l is the latent feature field at layer l;
$\mathcal{S}_l$ is the spectral branch;
$\mathcal{C}_l$ is the local convolution branch;
σ is the activation function.

Spectral transform

$$\hat{v}_l = \mathcal{F}(v_l), \qquad z_l = \mathcal{F}^{-1}(R_l \hat{v}_l)$$

where $\mathcal{F}$ is the Fourier transform;
$\mathcal{F}^{-1}$ is the inverse Fourier transform;
R_l is the trainable spectral weight tensor.

Interpretation:

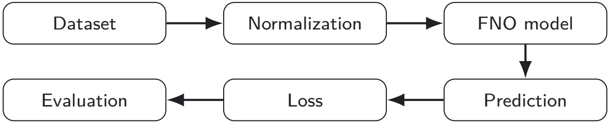
- Fourier transform moves the field into frequency space.
- Only low-frequency modes are retained and transformed.
- Inverse transform returns the result to physical space.
- Local convolution restores short-range spatial details.

Retained modes

$$\texttt{modes} = [K_1, K_2], \qquad d = 2$$

K_1, K_2 are the retained Fourier modes in two spatial directions.

Training and Evaluation of FNO



Training objective

$$\mathcal{L} = \lambda_{MSE}\mathcal{L}_{MSE} + \lambda_{rel}\mathcal{L}_{rel}$$

$\mathcal{L}$ is the total loss; $\mathcal{L}_{MSE}$ measures point-wise field error; $\mathcal{L}_{rel}$ measures relative L^2 deviation; λ_{MSE}, λ_{rel} control their contribution.

Normalization is computed on the training set and reused during validation, testing, and inference:

$$\tilde{s}_{i,c} = \frac{s_{i,c} - \mu_c}{\sigma_c}.$$

Evaluation includes:

- field-level visual comparison;
- channel-wise prediction errors;
- relative L^2 error;
- physical sanity checks.

Current caveat

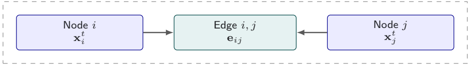
The FNO prototype is useful as a field-based neural-operator baseline, but still requires output-scale recalibration before being treated as a physically reliable predictor.

MeshGraphNet: Graph Input and Prediction

Main idea

MeshGraphNet represents the COMSOL mesh as a graph and predicts the next thermoelastic state at each mesh node.

Local graph connection



Graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

$\mathcal{G}$ – graph representation; $\mathcal{V}$ – mesh nodes; $\mathcal{E}$ – edges between neighbouring nodes.

Node features

$$\mathbf{x}_i^t = (\mathbf{p}_i, \mathbf{s}_i^t, \mathbf{m}_i, \mathbf{c})$$

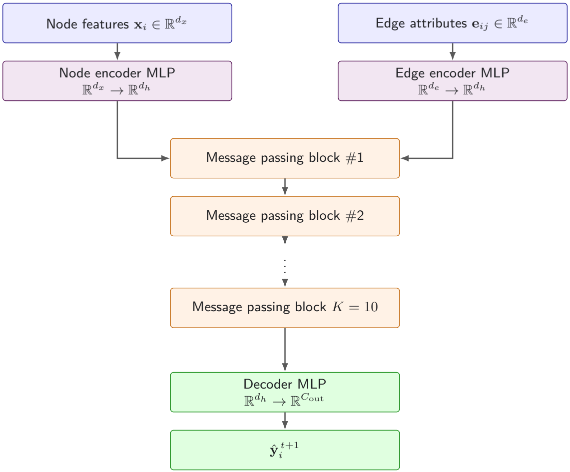
$\mathbf{x}_i^t$ – feature vector of node i at time t; $\mathbf{p}_i$ – node position; $\mathbf{s}_i^t$ – current physical state; $\mathbf{m}_i$ – material parameters; $\mathbf{c}$ – scenario condition.

Edge features

$$\mathbf{e}_{ij} = [\mathbf{p}_j - \mathbf{p}_i, \|\mathbf{p}_j - \mathbf{p}_i\|]$$

$\mathbf{e}_{ij}$ – edge attribute between nodes i and j; $\mathbf{p}_j - \mathbf{p}_i$ – relative position; $\|\mathbf{p}_j - \mathbf{p}_i\|$ – distance between nodes.

MeshGraphNet: Encoder–Processor–Decoder



The model uses latent dimension $d_h = 128$, ten message-passing steps, MLP encoders, and a node-wise decoder.

MeshGraphNet: Encoder and Decoder

Encoder

$$\mathbf{h}_i^0 = \phi_v^{enc}(\mathbf{x}_i^t), \qquad \mathbf{g}_{ij}^0 = \phi_e^{enc}(\mathbf{e}_{ij})$$

The encoder transforms raw node and edge information into latent vectors.

$\mathbf{x}_i^t$ is the node feature vector, $\mathbf{e}_{ij}$ is the edge feature vector, $\mathbf{h}_i^0$ is the initial latent node embedding, $\mathbf{g}_{ij}^0$ is the initial latent edge embedding.

Decoder

$$\hat{\mathbf{y}}_i^{t+1} = \phi^{dec}(\mathbf{h}_i^K)$$

The decoder converts the final latent node embedding into predicted physical fields for the next time step.

$\mathbf{h}_i^K$ is the node embedding after K message-passing steps, $\hat{\mathbf{y}}_i^{t+1}$ is the predicted output vector.

Implementation detail

Both encoders and the decoder are implemented as MLPs. In the current MeshGraphNet configuration, the latent dimension is $d_h = 128$, and the processor applies $K = 10$ message-passing steps.

MeshGraphNet: Message Passing Block

Edge update

$$\mathbf{e}'_{ij} = \mathrm{LN}\left(\mathbf{e}_{ij} + \phi_e([\mathbf{h}_i, \mathbf{h}_j, \mathbf{e}_{ij}])\right)$$

Aggregation

$$\mathbf{a}_j = \sum_{i:(i,j)\in\mathcal{E}} \mathbf{e}'_{ij}$$

Incoming edge messages are accumulated for every destination node using scatter-add aggregation.

Node update

$$\mathbf{h}'_j = \mathrm{LN}\left(\mathbf{h}_j + \phi_v([\mathbf{h}_j, \mathbf{a}_j])\right)$$

The residual connection preserves the previous node state and improves stability during repeated message passing.

Here ϕ_e and ϕ_v are MLPs with Linear layers, LayerNorm,3 SiLU activation, and optional Dropout.

Transformer: Inputs and Outputs

Input set (mesh nodes, state at $t = 0$)

$$a_1 = (x_1,\, T_1^{(0)},\, u_1^{(0)},\, \dot{u}_1^{(0)},\, a_1^{\mathsf{mat}})$$

$$a_2 = (x_2,\, \ldots)$$

$$\vdots$$

$$a_{N_{\text{in}}} = (x_{N_{\text{in}}},\, \ldots)$$

Query set (arbitrary points)

$$q_1 = (x_q^{(1)})$$

$$q_2 = (x_q^{(2)})$$

$$\vdots$$

$$q_{N_q} = (x_q^{(N_q)})$$

Input token $\in \mathbb{R}^{13}$

coordinates, initial state $(T, u, v, \dot{u}, \dot{v})$, and material parameters $(E, \nu, \rho, \alpha, k, C_p)$.

Query token $\in \mathbb{R}^{2}$

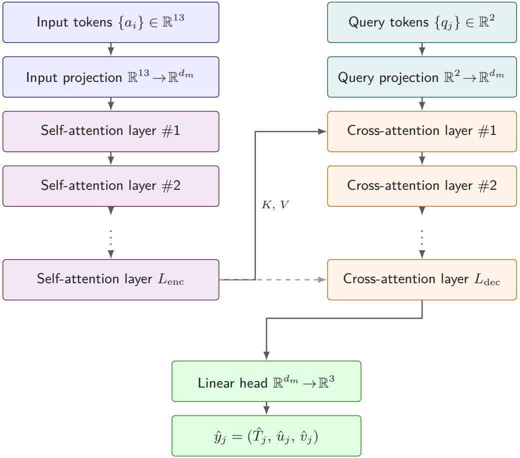
the coordinates $q_j = (x_q^{(j)}, y_q^{(j)})$ where fields are requested.

Output $\in \mathbb{R}^{3}$

$$\hat{y}_j = (\hat{T}_j,\, \hat{u}_j,\, \hat{v}_j)$$

N_{in} and N_q are independent, so the operator is discretisation-invariant.

Transformer: Architecture



Encoder self-attention; decoder cross-attention from queries; linear head returns (T, u, v).

Transformer: Training Loss

Composite objective

$$\mathcal{L}(\theta) = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{th}}\mathcal{L}_{\text{th}}, \qquad (\lambda_{\text{sup}}, \lambda_{\text{vel}}, \lambda_{\text{th}}) = (1.0,\ 0.25,\ 0.05)$$

Supervised (data)

$$\mathcal{L}_{\text{sup}} = \frac{1}{N_q C_{\text{out}}} \sum_{j,c} \left(\hat{y}_{j,c} - y_{j,c}\right)^2$$

Reconstructs the COMSOL reference fields.

Velocity consistency

$$\mathcal{L}_{\text{vel}} = \frac{1}{N_q} \sum_{j} \left\| \partial_t \hat{u}_j - \dot{u}_j^{\text{ref}} \right\|_2^2$$

Aligns the displacement rate with the exported velocity.

Thermal residual

$$\mathcal{L}_{\text{th}} = \frac{1}{N_q} \sum_{j} \left(\partial_t \hat{T}_j - \mathfrak{a}_j \nabla^2 \hat{T}_j \right)^2$$

Enforces diffusion with $\mathfrak{a}_j = k_j / (\rho_j C_{p,j})$.

Optimised with AdamW; tokens subsampled at $N_{\text{in}} = N_q = 1024$.

Calculation Status

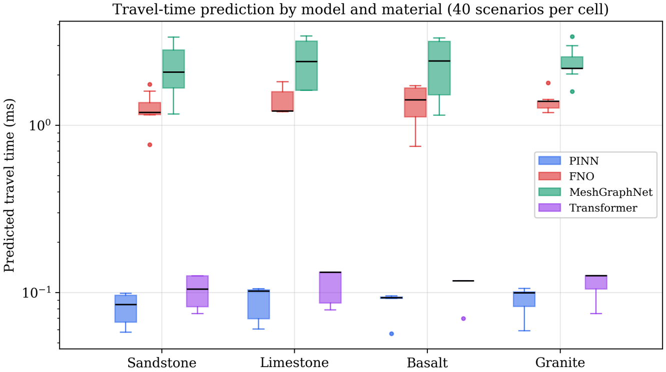
Calculation setup

1. **40 scenarios per material**, four materials: granite, sandstone, limestone, and basalt.
2. $40 \times 4 \times 4 = \mathbf{640}$ **model responses** evaluated under one shared contract.
3. **Zero fallback**: every plotted response is an active model output, not a mock.
4. **2D plane-strain only**: the operating envelope of all trained surrogates.

FNO caveat

FNO is included in the comparison, but its output scales are still miscalibrated. It should currently be treated as a diagnostic branch.

Travel-Time Predictions

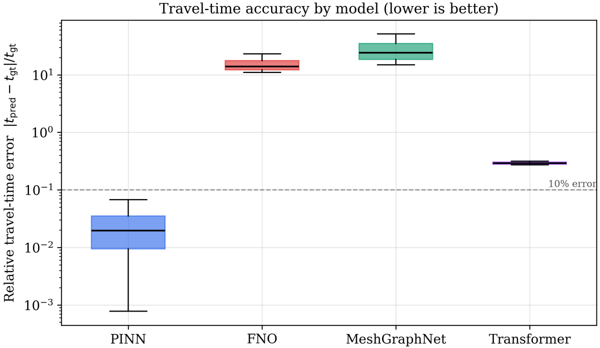


Reading the plot

- PINN ≈ 0.11 ms
- Transformer ≈ 0.14 ms
- FNO ≈ 1.29 ms
- MeshGraphNet ≈ 2.47 ms

Model-family differences are larger than the basalt–sandstone material difference in the current prototype.

Accuracy Against Analytical Reference



Analytical reference

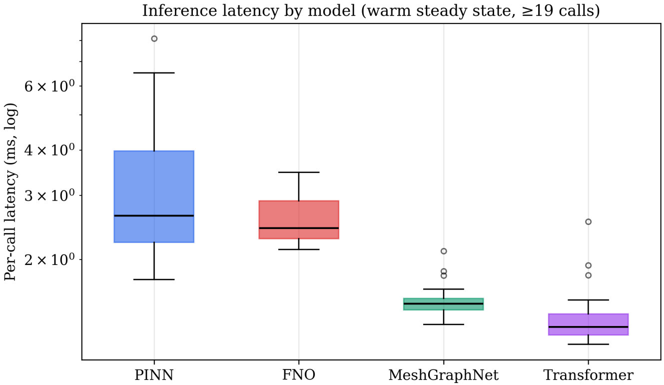
$$t_{gt} = \frac{r}{V_p}$$

t_{gt} – reference travel time [s]; r – source–probe distance [m]; V_p – compressional wave velocity [m/s].

Model	Median error	$\leq$10 %
PINN	2.0%	100%
Transformer	29.2%	0%
FNO	34.6%	0%
MeshGraphNet	$>$ 100%	0%

PINN is the only route that remains inside the 10% band across the balanced sweep.

Inference Latency and Time Behavior

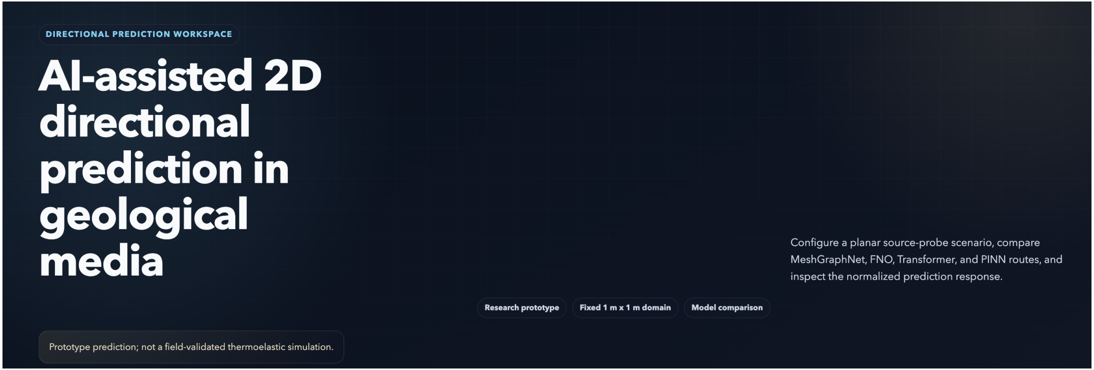


Median latency

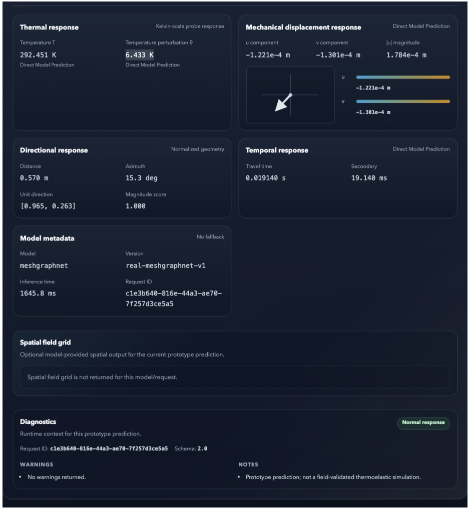
- Transformer ≈ 1.3 ms
- MeshGraphNet ≈ 1.5 ms
- FNO ≈ 2.4 ms
- PINN ≈ 2.6 ms

PINN is also the only route that changes smoothly with requested observation time; the other routes currently behave as fixed-horizon predictors.

Web Application Overview

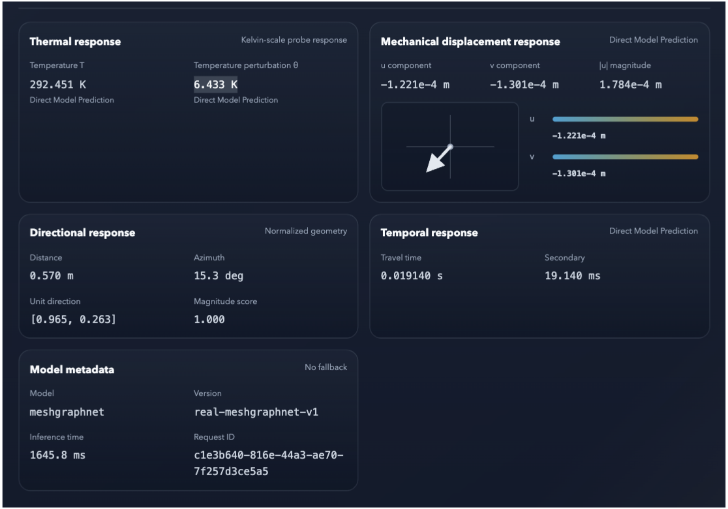


Directional Prediction Workspace

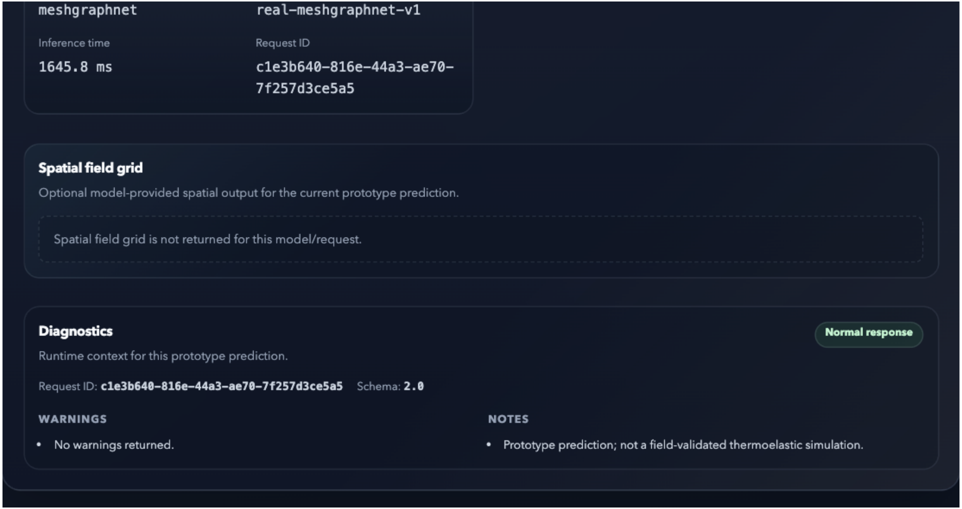


Source–Probe Visualization

Prediction Response Dashboard



Runtime Diagnostics and Metadata



Conclusions

1. The thesis develops a physically grounded *comparative framework* for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media.
2. PINN is the strongest *physics-informed baseline* in the current comparison and shows the best agreement with the analytical travel-time reference.
3. Transformer is the fastest route, while MeshGraphNet remains a useful mesh-aware surrogate candidate.
4. FNO should currently be treated as a diagnostic branch until output normalization is corrected.

Future Work

1. Retrain and recalibrate the 2D FNO route.
2. Add time-conditioned rollout behavior to MeshGraphNet.
3. Extend the material catalog with more complete thermoelastic parameters and validation data.
4. Compare against additional analytical or numerical baselines.

Final message

The prototype estimates directional characteristics of thermoelastic wave propagation within the simplified assumptions of the study.

Thank You

Questions?

Backup: Mechanical Rock Parameters

Material	**Type**	ρ kg/m^3	E GPa	ν –	μ GPa	K GPa	V_P/V_S m/s
Granite	intrusive	2650	76.28	0.245	30.63	49.84	5850/3400
Granodiorite	intrusive	2795	85.91	0.255	34.24	58.35	6100/3500
Basalt	volcanic	3000	103.96	0.266	41.07	73.95	6550/3700
Diabase	hypabyssal	2975	98.22	0.274	38.56	72.36	6450/3600
Gabbro	mafic	3075	114.27	0.287	44.40	89.33	6950/3800
Sandstone	sedimentary	2725	55.75	0.259	22.13	38.61	5000/2850
Limestone	carbonate	2675	61.48	0.277	24.08	45.90	5400/3000
Marble	metamorphic	2725	82.41	0.270	32.43	59.82	6150/3450
Schist	foliated	2800	68.20	0.267	26.91	48.82	5500/3100
Quartzite	quartz-rich	2650	83.50	0.250	33.40	55.70	6150/3550

Effective midpoint values under isotropic assumptions.

Backup: Thermal and Porosity Parameters

Material	α 10^{-6}/K	k W/(m K)	C_p J/(kg K)	C MJ/(m^3 K)	γ MPa/K	**Porosity** %
Granite	7.9	2.50	850	2.25	1.18	total 45.5
Granodiorite	–	2.40	–	–	–	–
Basalt	–	1.95	1300	3.90	–	total 19.0
Diabase	–	2.50	850	2.53	–	–
Gabbro	–	2.30	1000	3.08	–	total 43.5
Sandstone	11.6	2.25	1000	2.73	1.34	total 31.5; eff. 26.5
Limestone	8.0	2.55	1050	2.81	1.10	total 31.5; eff. 18.0
Marble	9.8	2.80	850	2.32	1.76	–
Schist	–	2.75	1150	3.22	–	total 26.5; eff. 27.5
Quartzite	–	5.10	1000	2.65	–	–

Dashes mark values that are not available in the source catalogue.

Backup: Elastic Wave Velocities

Compressional (P) waves

Particle motion is mainly parallel to the propagation direction and involves volume change.

Shear (S) waves

Particle motion is mainly transverse to the propagation direction and depends on shear rigidity.

$$V_P = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \qquad V_S = \sqrt{\frac{G}{\rho}}$$

V_P, V_S wave velocities [m/s]; K bulk modulus [Pa]; G shear modulus [Pa]; ρ density [kg/m^3]. Stiffness increases velocity; density increases inertia.

Backup: Thermal Diffusivity

Definition

$$\alpha_T = \frac{k}{\rho\, C_p}$$

α_T thermal diffusivity [m^2/s]. Larger α_T means a temperature disturbance spreads faster. This quantity controls the time-scale of the thermal part of the coupled response; it is reported here because the main flow only shows the heat equation in differential form.

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References I

- G. Mavko, T. Mukerji, and J. Dvorkin, *The Rock Physics Handbook: Tools for Seismic Analysis of Porous Media*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009.
- J. C. Jaeger, N. G. W. Cook, and R. W. Zimmerman, *Fundamentals of Rock Mechanics*, 4th ed. Malden, MA, USA: Blackwell, 2007.
- E. C. Robertson, *Thermal Properties of Rocks*. Reston, VA, USA: U.S. Geological Survey, Open-File Rep. 88–441, 1988.
- D. L. Turcotte and G. Schubert, *Geodynamics*, 3rd ed. Cambridge, U.K.: Cambridge Univ. Press, 2014.
- R. G. Wolff, *Physical Properties of Rocks: Porosity, Permeability, Distribution Coefficients, and Dispersivity*. Reston, VA, USA: U.S. Geological Survey, Open-File Rep. 82–166, 1982.
- K. Aki and P. G. Richards, *Quantitative Seismology*, 2nd ed. Sausalito, CA, USA: Univ. Science Books, 2002.
- J. E. Lucius, W. H. Langer, and K. J. Ellefsen, *An Introduction to Using Surface Geophysics to Characterize Sand and Gravel Deposits*. Reston, VA, USA: U.S. Geological Survey, Circular 1310, 2007.
- W. Nowacki, *Thermoelasticity*, 2nd ed. Oxford, U.K.: Pergamon Press; Warsaw, Poland: PWN–Polish Scientific Publishers, 1986.
- B. A. Boley and J. H. Weiner, *Theory of Thermal Stresses*. New York, NY, USA: Wiley, 1960.
- B. Alipova, “Distribution of coupled thermoelastic waves in different rocks using a MATLAB GUI application,” in *Proc. DTESI 2024: 9th Int. Conf. Digital Technologies in Education, Science and Industry*, CEUR Workshop Proceedings, vol. 3966, 2025. [Online]. Available: `https://ceur-ws.org/Vol-3966/W1Paper2.pdf`

[II

References]References

M. F. Ismail, H. M. Ahmed, S. Safwat, M. E. Ramadan, N. S. E. Abdalla, and M. Soliman, "Analytical wave solutions in thermoelastic media with temperature-dependent properties via the IMETF method," *Scientific Reports*, vol. 15, Art. no. 32553, 2025, doi: 10.1038/s41598-025-09344-w.

W. Hou, L.-Y. Fu, and J. M. Carcione, "Amplitude-variation-with-offset in thermoelastic media," *Geophysics*, vol. 88, no. 1, pp. MR25–MR33, 2023, doi: 10.1190/geo2021-0815.1.

B. Yang, Z. Li, L. Zeng, and X. Chen, "Simulation of thermoelastic wave propagation in 3-D multilayered half-space media," *Geophysical Journal International*, vol. 232, no. 2, pp. 1408–1426, 2023, doi: 10.1093/gji/ggac401.

S. Crampin, "A review of wave motion in anisotropic and cracked elastic-media," *Wave Motion*, vol. 3, no. 4, pp. 343–391, 1981.

L. Thomsen, "Weak elastic anisotropy," *Geophysics*, vol. 51, no. 10, pp. 1954–1966, 1986.

N. A. Ispulov, A. Zh. Zhumabekov, A. Qadir, A. A. Kurmanov, Sh. N. Sarymova, K. R. Dossumbekov, and E. Arinov, "The propagation of thermoelastic waves in different anisotropic media using the matricant method," *Advances in Mathematical Physics*, vol. 2022, Art. no. 5787899, 2022, doi: 10.1155/2022/5787899.

R. A. Schultz, *Geologic Fracture Mechanics*. Cambridge, U.K.: Cambridge Univ. Press, 2019.

F. H. Cornet, *Elements of Crustal Geomechanics*. Cambridge, U.K.: Cambridge Univ. Press, 2007.

S. Earle, *Physical Geology*, 2nd ed. Victoria, BC, Canada: BCcampus Open Education, 2019. [Online]. Available: `https://opentextbc.ca/physicalgeology2ed/`

[III

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C. H. Moore, *Carbonate Reservoirs: Porosity, Evolution and Diagenesis in a Sequence Stratigraphic Framework*. Amsterdam, The Netherlands: Elsevier, 2001.

W. Moebs, S. J. Ling, and J. Sanny, *University Physics, Volume 1, Chapter 16: Waves*. Houston, TX, USA: OpenStax, 2016. [Online]. Available: https://assets.openstax.org/oscms-prodcms/media/documents/UniversityPhysicsVol1-WEB.pdf

R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics, Vol. I, Ch. 51: Waves*. Pasadena, CA, USA: California Institute of Technology. [Online]. Available: https://www.feynmanlectures.caltech.edu/I_51.html

M. P. Volarovich and E. I. Bayuk, *Fizicheskie svoistva mineralov i gornykh porod pri vysokikh davleniyakh i temperaturakh*. Moscow, Russia: Nedra, 1988.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998–6008, 2017.

Z. Li, K. Meidani, and A. B. Farimani, "Transformer for partial differential equations' operator learning," *Transactions on Machine Learning Research*, 2023.

S. Cao, "Choose a transformer: Fourier or Galerkin," *Advances in Neural Information Processing Systems*, vol. 34, pp. 24924–24940, 2021.

H. Wu, H. Luo, H. Wang, J. Wang, and M. Long, "Transolver: A fast transformer solver for PDEs on general geometries," in *Proc. 41st Int. Conf. Machine Learning*, 2024.